

Name
Date
Course

Rational Functions and Their Graphs (Homework)

Graph the following functions. Draw asymptotes as dotted lines and write their equations.

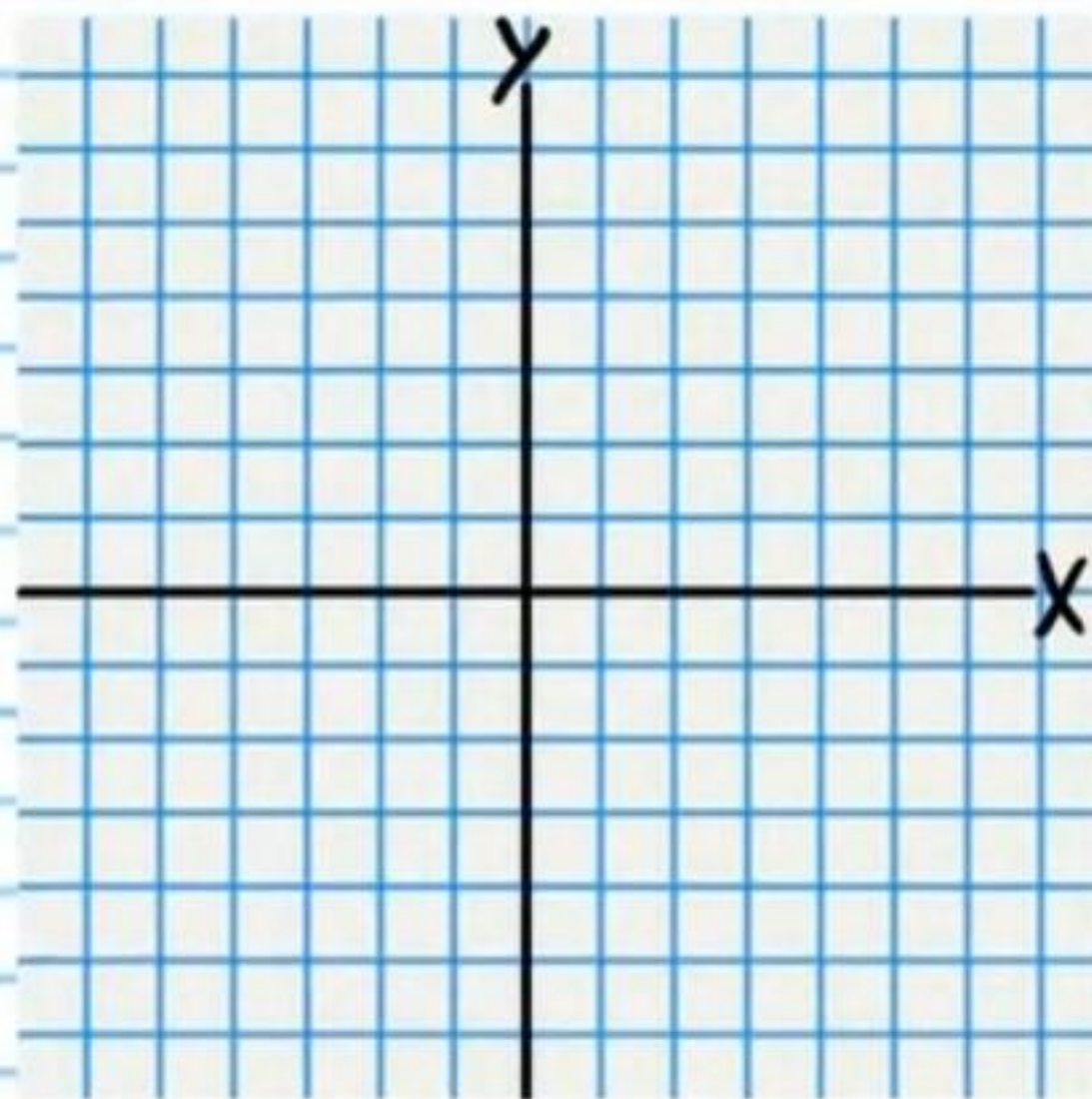
① $f(x) = \frac{1}{x^2 - 6x + 9}$ Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



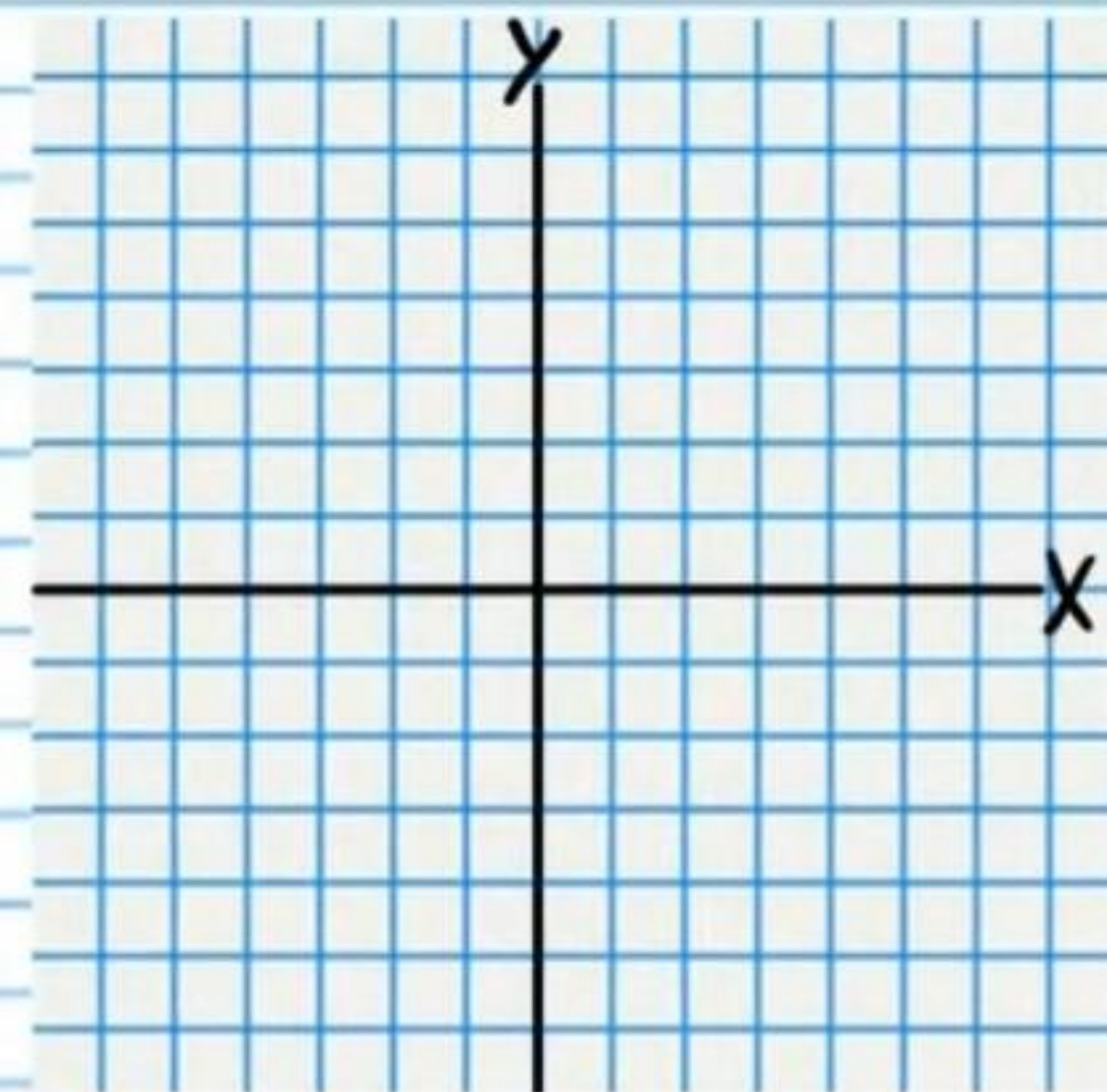
② $f(x) = \frac{2}{x^2 - x - 20}$ Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



③ $f(x) = \frac{3x+3}{x^2+3x-4}$

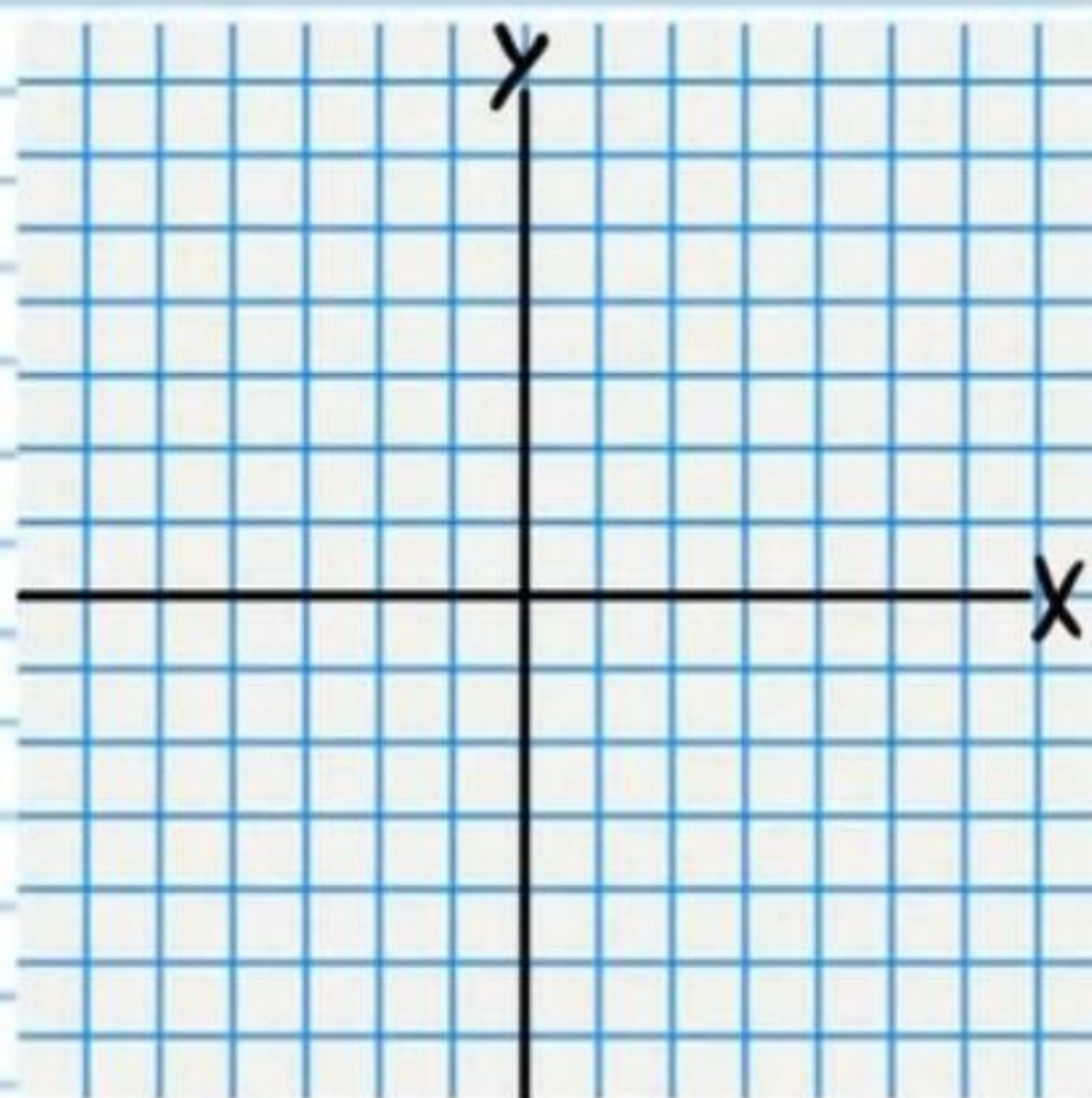
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



④ $f(x) = \frac{x-1}{x^2-x-6}$

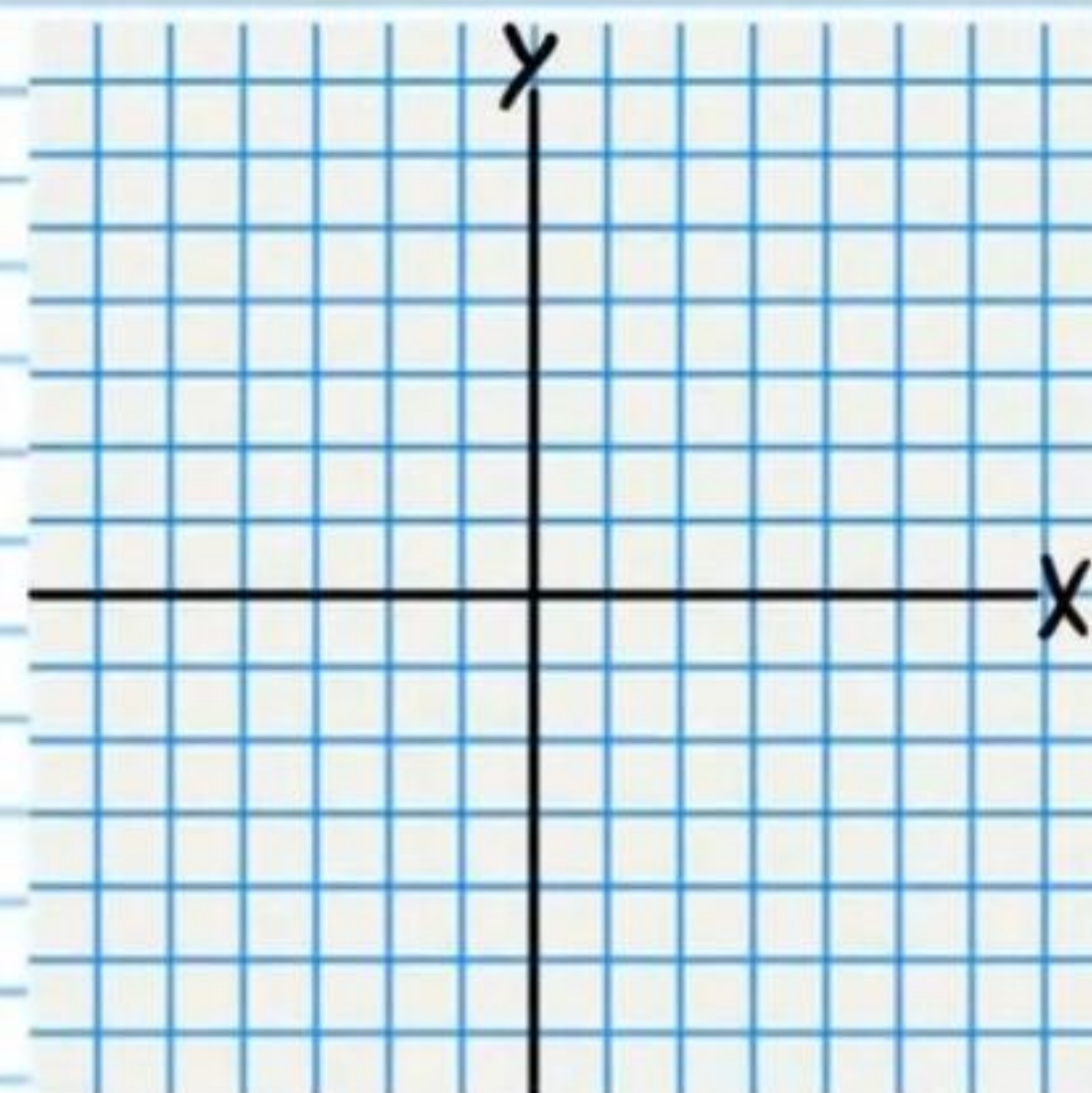
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



* Write the coordinates of holes and draw them as little circles.

$$⑤ f(x) = \frac{x-1}{x^2-3x+2}$$

Hole

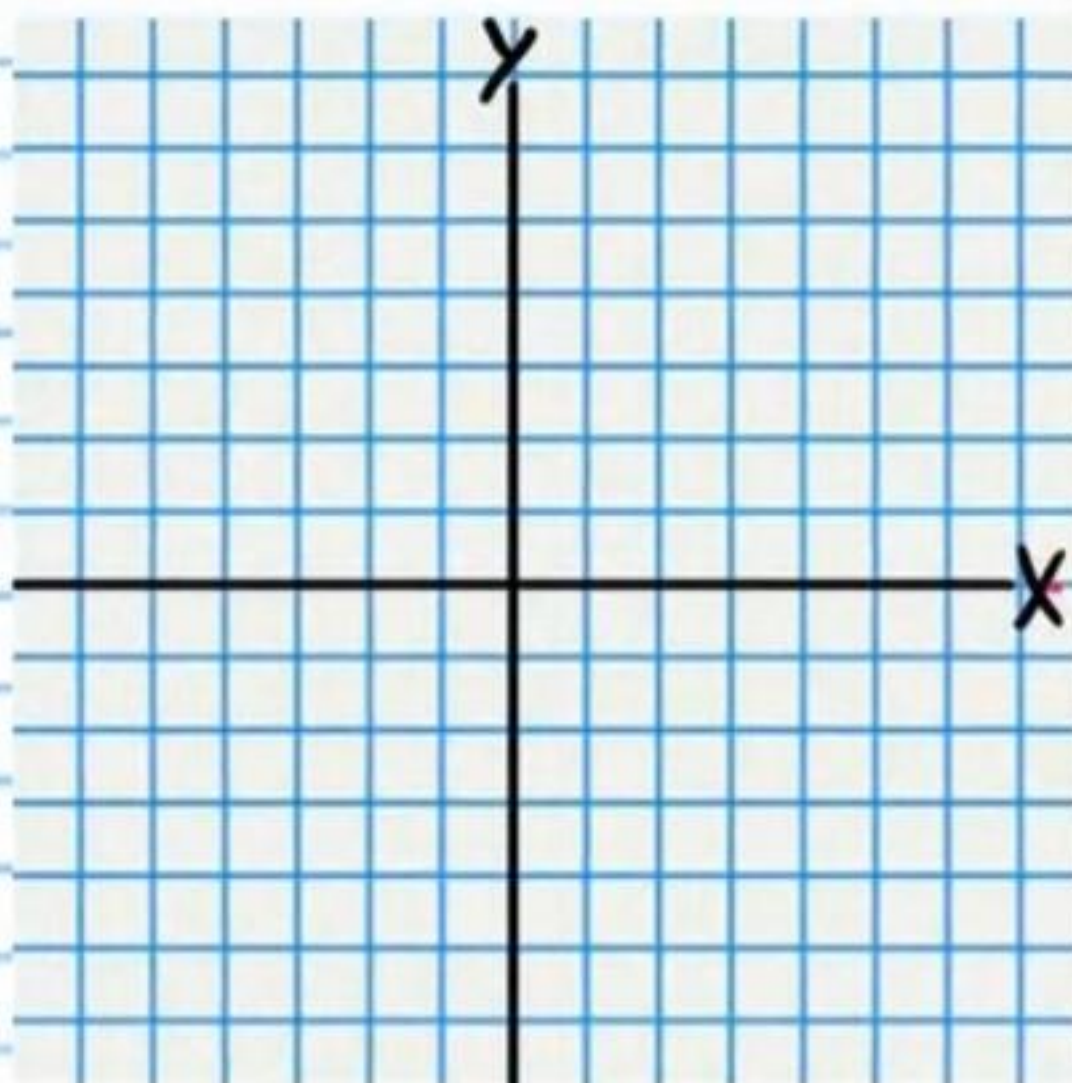
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



$$⑥ f(x) = -\frac{2x+8}{x^2+5x+4}$$

Hole

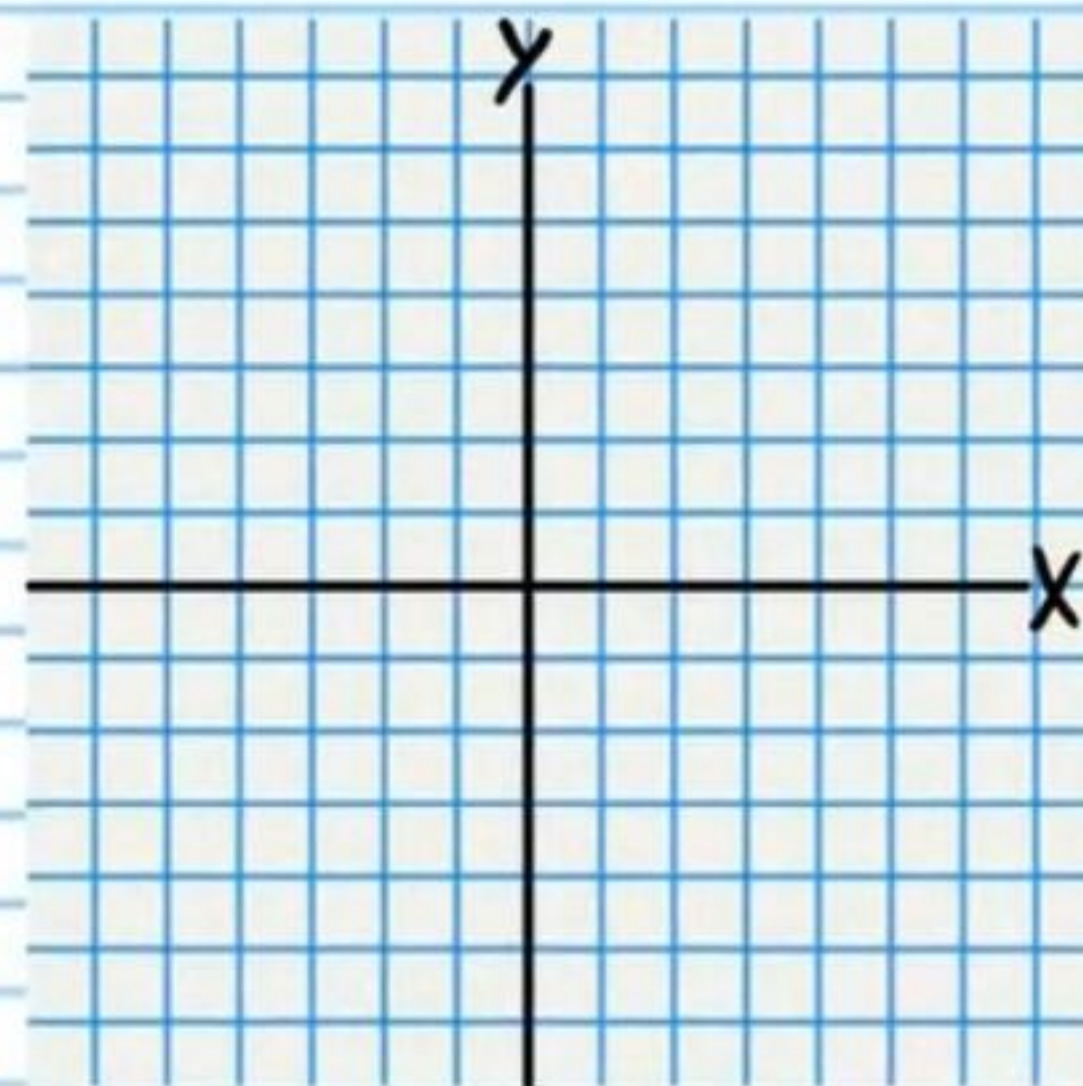
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



Answers

① $f(x) = \frac{1}{x^2 - 6x + 9}$

Vertical Asymptote Equations

$x = 3$

Horizontal Asymptote Equations

$y = 0$

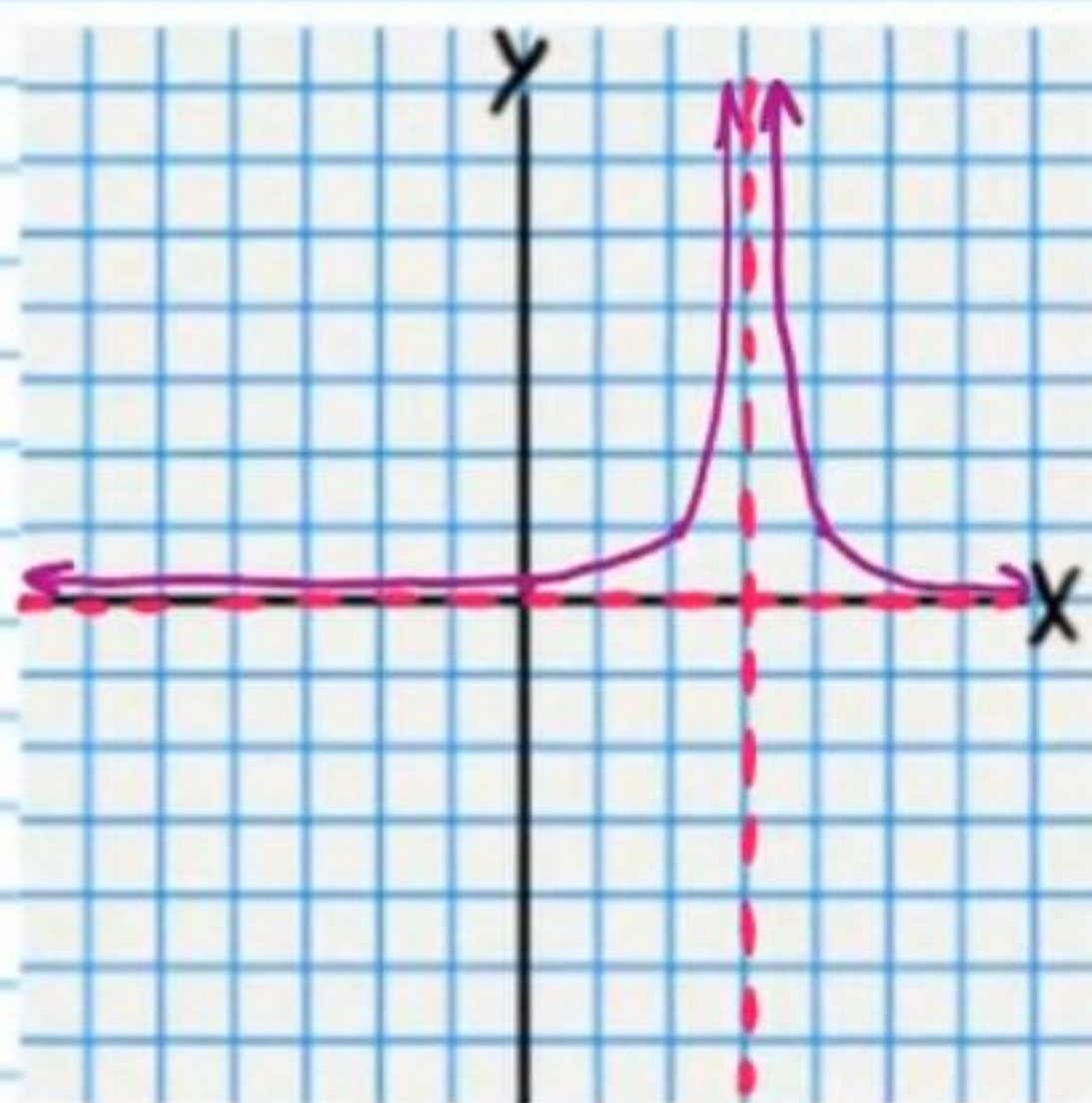
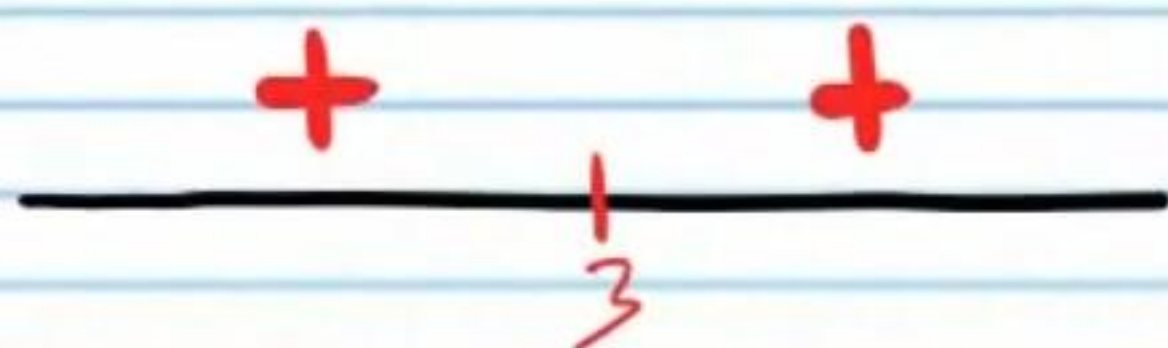
Sign Chart

Y-Intercept

$(0, \frac{1}{9})$

X-Intercept

None



② $f(x) = \frac{2}{x^2 - x - 20}$

Vertical Asymptote Equations

$x = 5, x = -4$

Horizontal Asymptote Equations

$y = 0$

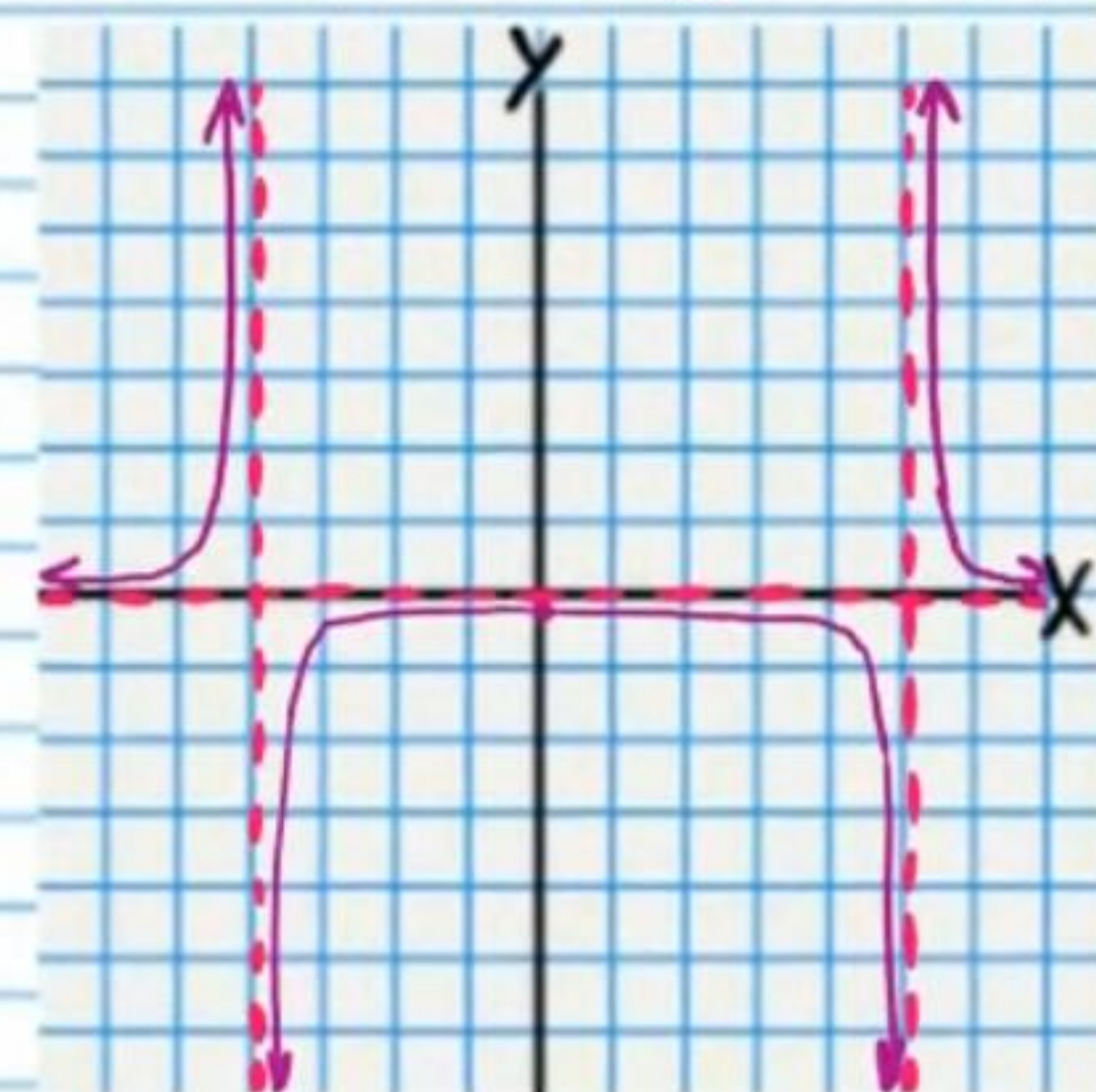
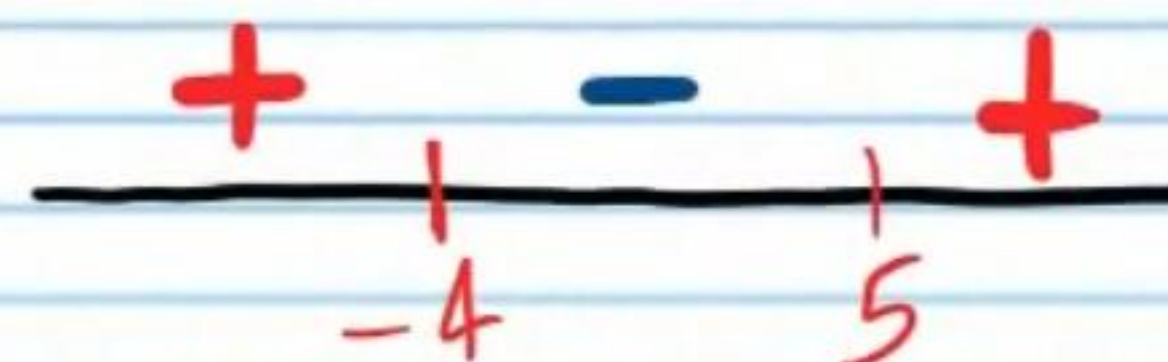
Sign Chart

Y-Intercept

$(0, -\frac{1}{10})$

X-Intercept

None



$$\textcircled{3} f(x) = \frac{3x+3}{x^2+3x-4}$$

Vertical Asymptote Equations

$$x = -4, x = 1$$

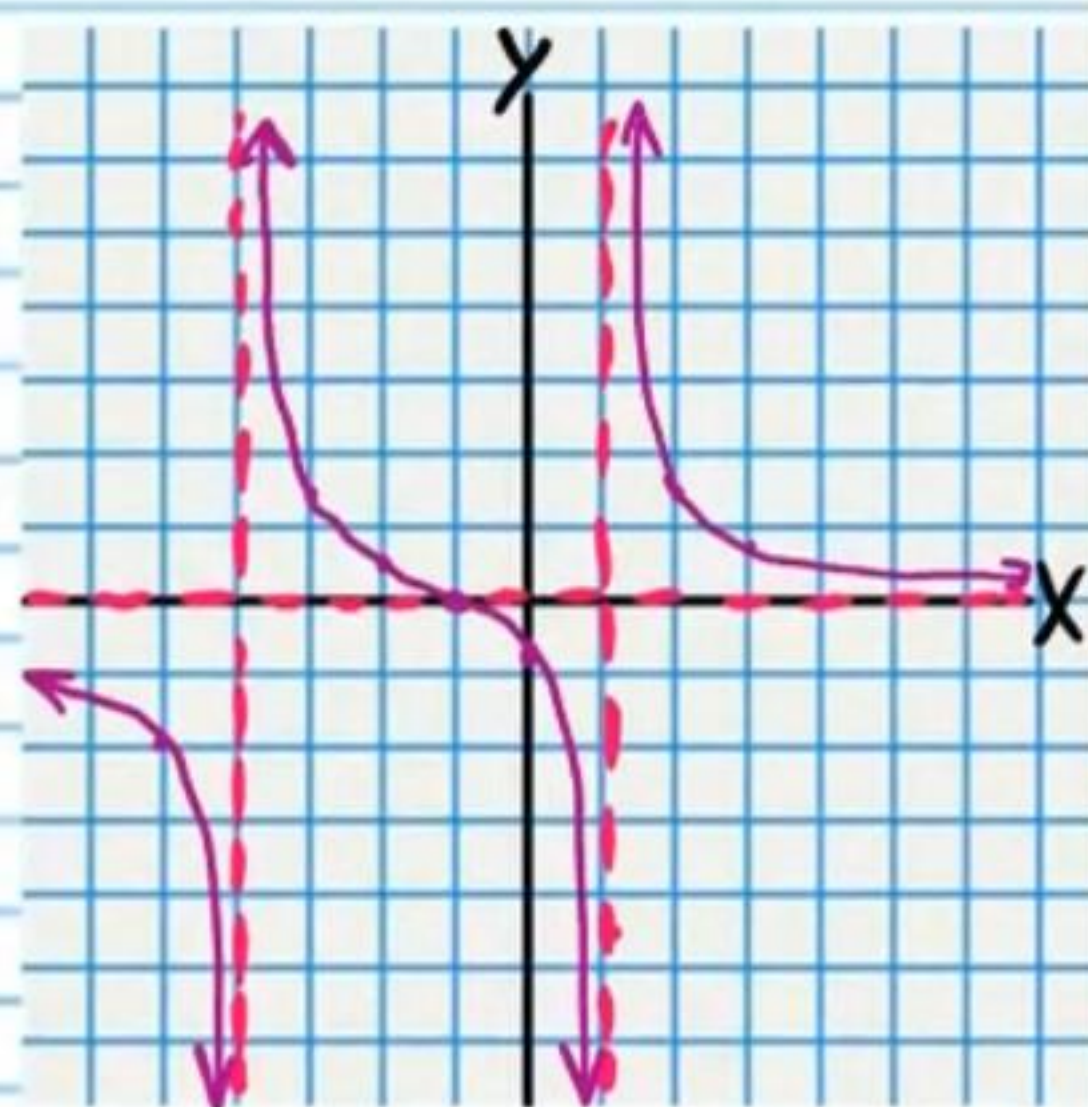
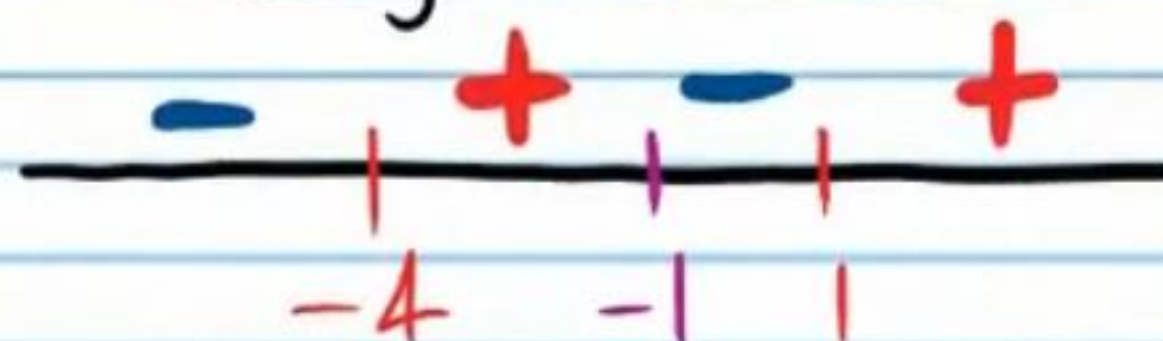
Horizontal Asymptote Equations

$$y = 0$$

Y-Intercept

$$(0, -3/4)$$

Sign Chart



X-Intercept

$$(-1, 0)$$

$$\textcircled{4} f(x) = \frac{x-1}{x^2-x-6}$$

Vertical Asymptote Equations

$$x = 3, x = -2$$

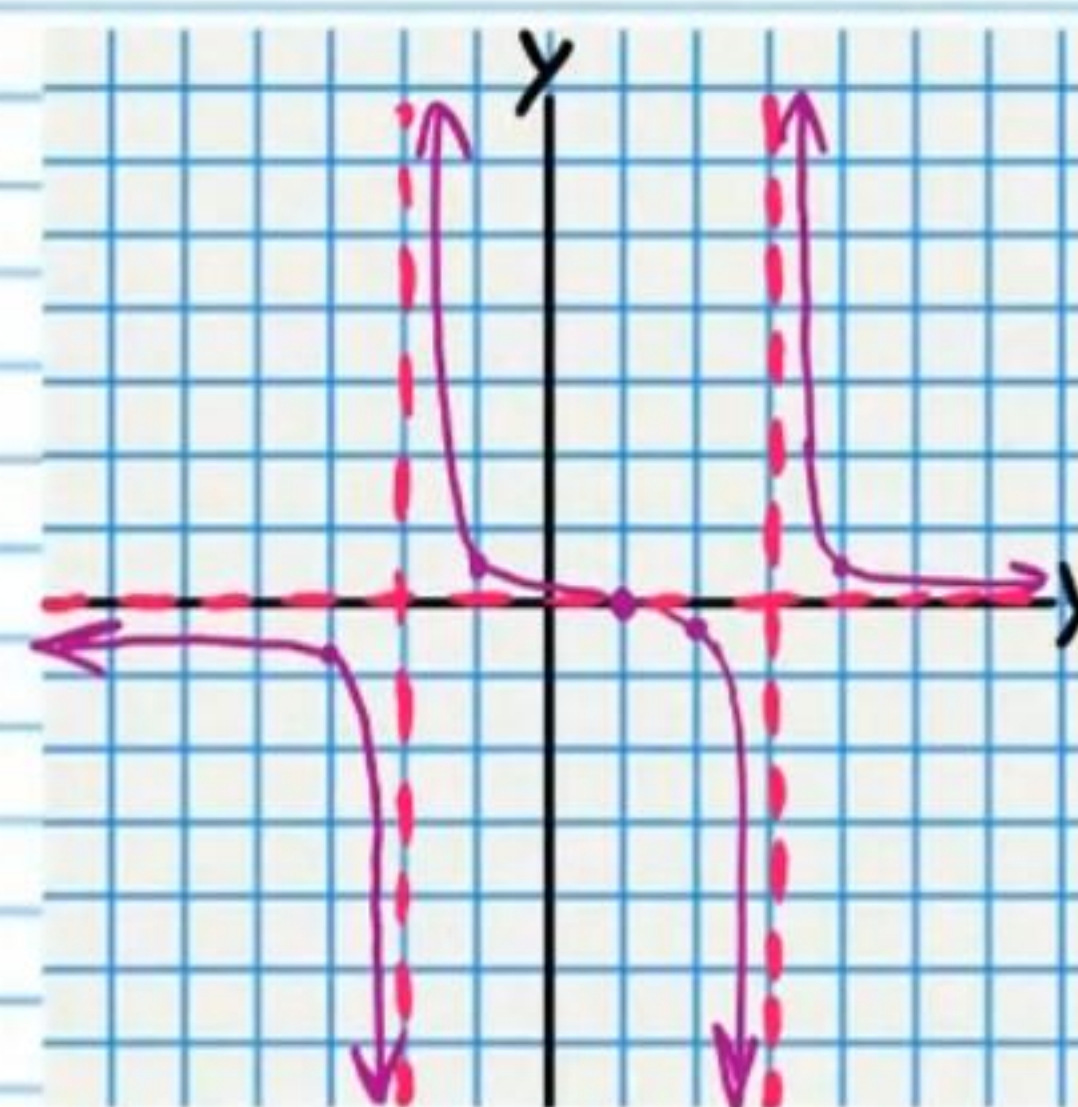
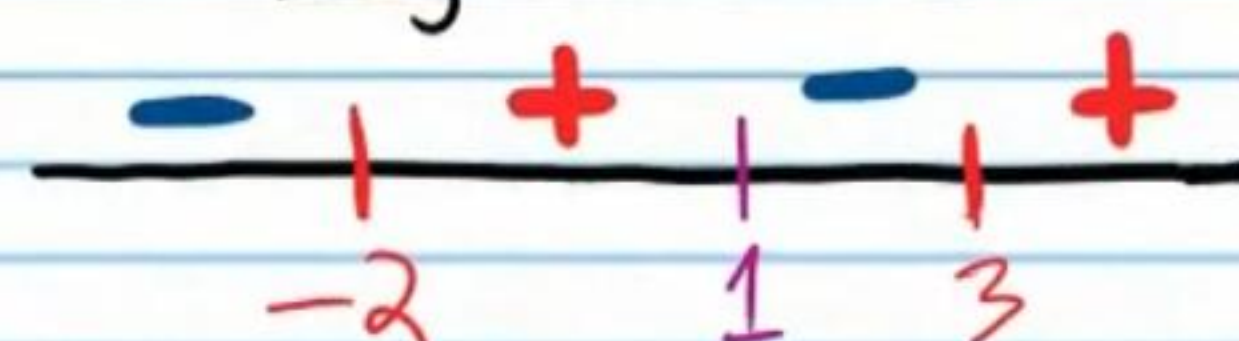
Horizontal Asymptote Equations

$$y = 0$$

Y-Intercept

$$(0, 1/6)$$

Sign Chart



X-Intercept

$$(1, 0)$$

$$⑤ f(x) = \frac{x-1}{x^2-3x+2}$$

Hole

$$x=1$$

$$g(1) = -1$$

$$(1, -1)$$

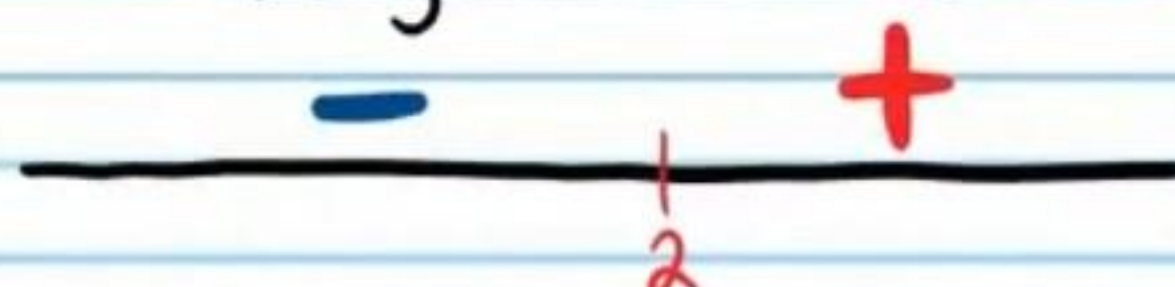
Vertical Asymptote Equations

$$x=2$$

Horizontal Asymptote Equations

$$y=0$$

Sign Chart

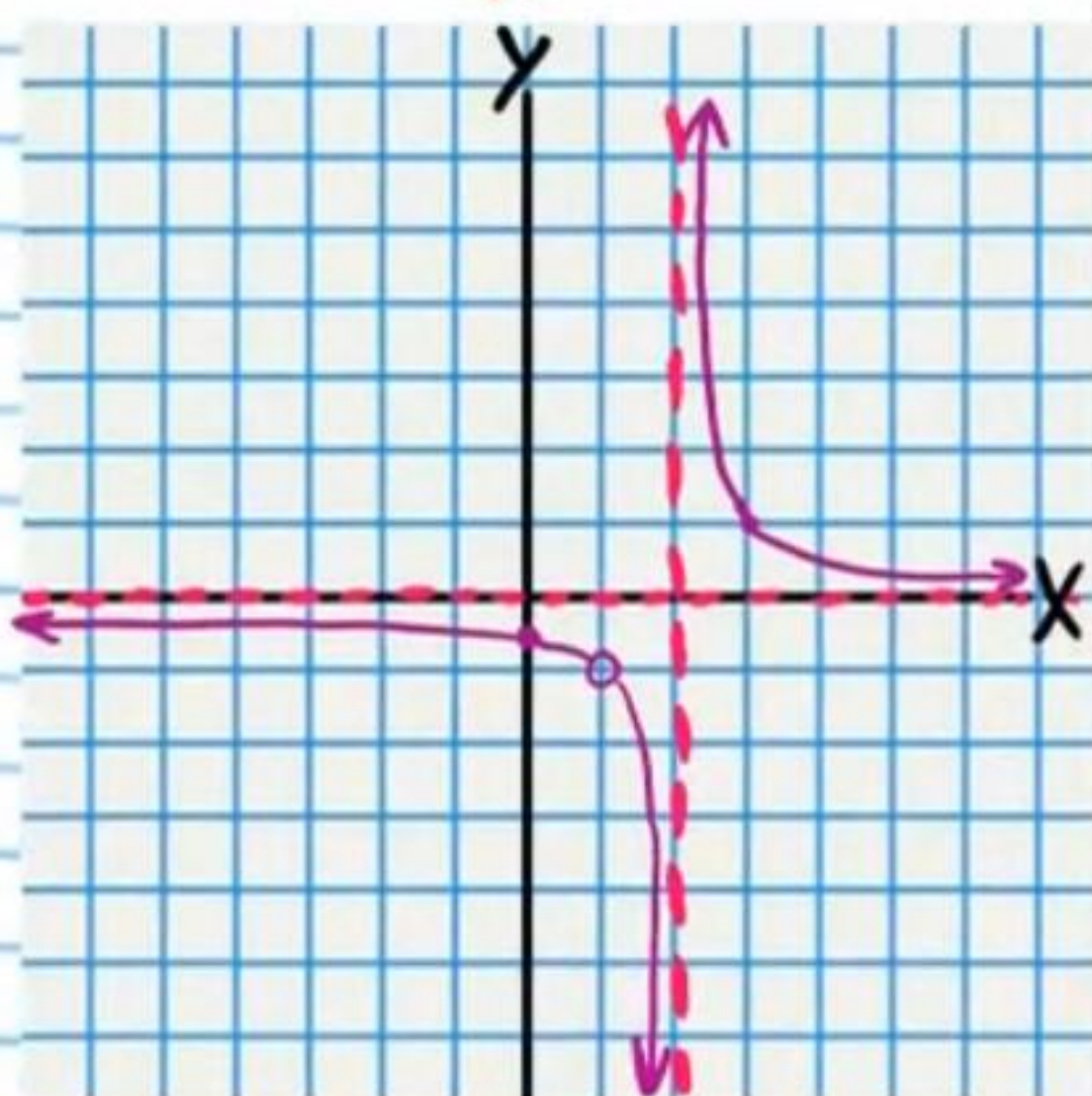


Y-Intercept

$$(0, -\frac{1}{2})$$

X-Intercept

None



$$⑥ f(x) = -\frac{2x+8}{x^2+5x+4}$$

Hole

$$x=-4$$

$$g(-4) = \frac{2}{3}$$

$$(-4, \frac{2}{3})$$

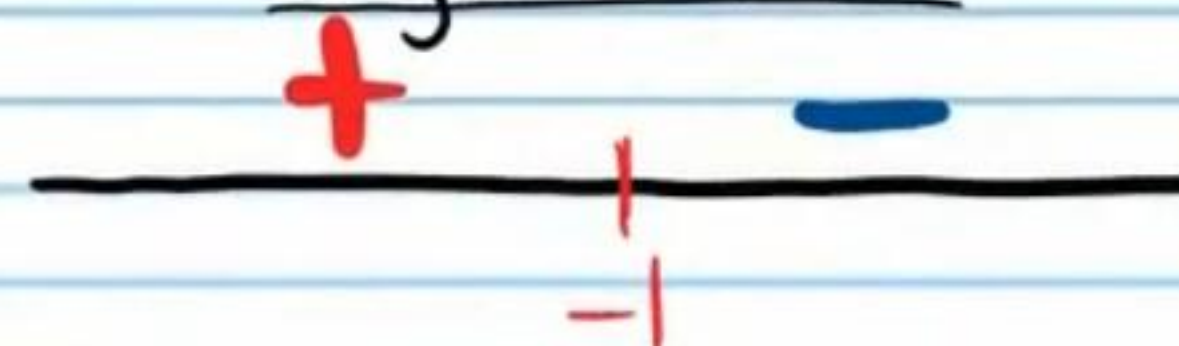
Vertical Asymptote Equations

$$x=-1$$

Horizontal Asymptote Equations

$$y=0$$

Sign Chart



Y-Intercept

$$(0, -2)$$

X-Intercept

None

